A Modular Quantitative Trading Pipeline

Alpha Generation, Reinforcement Learning Control, and Risk-Aware Backtesting

Sequential Decision-Making Project

Ioan Hedea

March 23, 2026

Executive summary. The pipeline combines cross-sectional factor investing, statistical arbitrage, volatility and regime modeling, and a lightweight recurrent forecast sleeve with reinforcement-learning-based portfolio sizing and dynamic hedging. The current implementation is modularized under `stock_trading/quant_stack` and preserves a thin legacy entrypoint for reproducible runs.

Abstract

This report documents a modular quantitative trading pipeline implemented in the `quant_stack` package under `stock_trading`. The system combines cross-sectional factor signals, statistical arbitrage, volatility forecasting, regime detection, and a lightweight recurrent forecast sleeve with reinforcement-learning-based portfolio allocation, hedging, and execution demonstrations. The resulting design is intentionally hybrid: classical quantitative finance supplies the alpha and risk features, while reinforcement learning is used where it is most natural, namely dynamic sizing, exposure control, and sequential decision-making under changing market conditions.

1 Objective

The goal of the project is not simply to outperform a benchmark on raw return. The stronger objective is to construct a research pipeline that is:

- broad enough to resemble a multi-sleeve quantitative strategy,
- modular enough to be maintained and extended cleanly,
- financially grounded rather than purely heuristic,
- explicit about where reinforcement learning helps and where it does not.

In this design, the alpha layer identifies opportunities, while the RL layer decides how aggressively to express them and how to protect the portfolio in stress.

2 Architecture

At a high level, the system is organized as

$$\begin{aligned}\text{Data} &\to \text{Features} \to \text{Alpha Models}\\ &\to \text{Adaptive Combiner} \to \text{Portfolio RL}\\ &\to \text{Hedging / Cash} \to \text{Backtest and Diagnostics.}\end{aligned}$$

2.1 Code Structure

The implementation is separated into the following modules:

Module	Responsibility
`config.py`	Universe, constants, and external series identifiers
`data.py`	Market, macro, and SEC data loading
`alpha.py`	Factor, pairs, GARCH, HMM, LSTM, and signal combination
`rl.py`	Portfolio construction, execution, and hedge RL agents
`pipeline.py`	Walk-forward backtest orchestration
`plots.py`	Performance, diagnostics, and execution figures
`main.py`	CLI entrypoint used by the wrapper script

This separation keeps the external entrypoint stable while making the internal system easier to reason about and extend.

3 Data Layer

3.1 Assets

The equity and diversifier universe spans growth, financials, energy, defensives, and cross-asset hedging proxies:

- Mega-cap technology: AAPL, MSFT, GOOGL, AMZN, META, NVDA
- Financials and energy: JPM, GS, XOM, CVX
- Defensives and staples: JNJ, PG, KO, PEP, WMT
- Diversifiers: GLD, TLT, XLU, VNQ
- Benchmark: SPY

3.2 Sources

The pipeline supports several data channels:

- **Prices and volumes**: `yfinance`
- **Macro, preferred**: FRED when `FRED_API_KEY` is present
- **Macro, no-key fallback**: BLS plus U.S. Treasury Fiscal Data
- **Fundamentals**: SEC company facts when `SEC_USER_AGENT` is set

This setup intentionally supports both a zero-cost mode and a richer research mode with additional macro and filing-based information.

3.3 Macro Regime Features

When available, the macro sleeve includes:

- 10-year Treasury yield
- 2-year Treasury yield
- Federal funds rate
- unemployment rate
- term spread

These features do not act as a stand-alone alpha engine. Instead, they inform a macro regime belief that is blended with the HMM market-regime estimate.

4 Alpha Layer

4.1 Cross-Sectional Factor Composite

The factor model forms a composite score from four standard sleeves:

- momentum,
- value proxy,
- quality proxy,
- low volatility.

If m_i, v_i, q_i, ℓ_i denote z-scored factor values for asset i, the composite factor score is

$$f_i = 0.30m_i + 0.25v_i + 0.25q_i + 0.20\ell_i.$$

The quality term uses return-stability information and, when available, a simple SEC-based quality score blended into the cross-section.

4.2 Pairs Trading

Pairs trading follows the classical Engle-Granger idea:

1. test candidate pairs for cointegration,
2. estimate a hedge ratio via OLS,
3. compute the spread,
4. trade deviations from the equilibrium spread.

For a pair $(P_t^{(1)}, P_t^{(2)})$ with hedge ratio β, the spread is

$$s_t = P_t^{(1)} - \beta P_t^{(2)}.$$

Its z-score is

$$z_t = \frac{s_t - \mu_s}{\sigma_s + \varepsilon}.$$

Large positive z-scores imply short-spread trades, while large negative z-scores imply long-spread trades.

4.3 Volatility Forecasting

Each name receives a GARCH(1,1) volatility forecast of the form

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

These forecasts serve two roles:

- they provide a confidence scaling for signal sizing,
- they help the control layer understand when apparent alpha should be discounted because the environment is too volatile.

4.4 Regime Detection

The market regime sleeve uses a simple two-state HMM on market returns. The states correspond loosely to bullish and bearish regimes. The HMM belief is then blended with a macro regime signal:

$$\text{RegimeBelief}_t = 0.75 \cdot \text{HMMBelief}_t + 0.25 \cdot \text{MacroBelief}_t.$$

This blended belief is later used to adjust both signal strength and RL state.

4.5 LSTM Sleeve

A lightweight recurrent model is trained on rolling windows of engineered return features, including short-horizon momentum, rolling volatility, and normalized returns. The purpose is not to create a heavy deep-learning stack, but to add a nonlinear, time-series-aware signal sleeve that can be adaptively up- or down-weighted.

4.6 Adaptive Signal Combination

Let $s_{i,t}^{(k)}$ denote the score for asset i from source k. The combiner forms

$$\alpha_{i,t} = \sum_k w_t^{(k)} s_{i,t}^{(k)},$$

where the source weights $w_t^{(k)}$ are adjusted using recent realized signal quality. In the implementation, realized skill is proxied by recent Spearman information coefficients between source scores and realized returns. This makes the alpha stack adaptive rather than statically weighted.

5 Reinforcement Learning Layer

5.1 Why RL is Used Here

The project uses RL where sequential control is natural:

- deciding how much capital to allocate,
- deciding when to hedge more aggressively,
- deciding how to split execution in the stand-alone demo.

This is a more defensible use of RL than asking it to learn stock-selection alpha directly from scratch in a small-data setting.

5.2 Portfolio Construction RL

The portfolio RL state is based on:

- cross-sectional alpha opportunity,
- current realized portfolio volatility,
- regime belief.

Actions choose an exposure bucket and an active-overlay size. The current implementation uses a factor-anchored target book:

$$w_t^{\text{target}} = 0.90\, w_t^{\text{factor}} + 0.10\, w_t^{\text{stabilizer}},$$

where the stabilizer is a long-only minimum-variance style portfolio estimated from a Ledoit-Wolf shrinkage covariance matrix.

If the RL action implies invested fraction b_t and overlay fraction τ_t, the final invested weights are approximately

$$w_t = b_t \left[(1 - \tau_t) w_t^{\text{target}} + \tau_t w_t^{\text{satellite}} \right].$$

The satellite sleeve is not an entirely independent portfolio. It tilts the factor target toward the strongest alpha names, which keeps the strategy close to the factor book while still allowing active expression.

5.3 No-Trade Band

Small reallocations are suppressed by a rebalance band. If the implied turnover is too small, the system keeps the previous portfolio. If only some positions are inside the deadband, those positions remain unchanged and the rest of the book is rescaled to preserve the target capital budget. This is meant to reduce turnover drag without silently changing total exposure.

5.4 Hedging RL

The hedging agent uses state variables based on:

- current drawdown,
- volatility regime percentile,
- recent momentum.

It selects one of a small number of hedge ratios. The hedge sleeve is modeled as a convex, options-like protection layer:

$$r_t^{\text{total}} = r_t^{\text{risky}}(1 - \phi h_t) + r_t^{\text{cash}} + r_t^{\text{hedge}},$$

where h_t is the hedge intensity, ϕ discounts the risky sleeve under protection, and the hedge sleeve includes both a carry cost and a convex bonus in stress.

5.5 Execution RL

The execution component is a separate demonstration rather than the main driver of backtest results. It learns how to split a large order over a stylized trading day to reduce market impact relative to naive or schedule-based execution.

6 Backtesting Protocol

The backtest is walk-forward:

- historical prices and volumes are loaded,
- the sample is split into a training half and a test half,
- LSTM sleeves are trained on the initial window,
- the pipeline then steps forward day by day through the test period.

At each step, the system:

1. computes factor, pairs, GARCH, HMM, macro, and LSTM features,
2. combines them into expected returns and confidence scores,
3. lets RL choose exposure and hedge actions,
4. applies transaction costs and cash carry,
5. updates RL values from realized outcomes.

7 Evaluation Metrics

The reporting suite includes:

- annualized return,
- annualized volatility,
- Sharpe ratio,
- Sortino ratio,
- maximum drawdown,
- Calmar ratio,
- return distribution diagnostics including daily VaR.

For example, the annualized Sharpe ratio is computed as

$$\text{Sharpe} = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}} + \varepsilon}.$$

Similarly, the Calmar ratio is

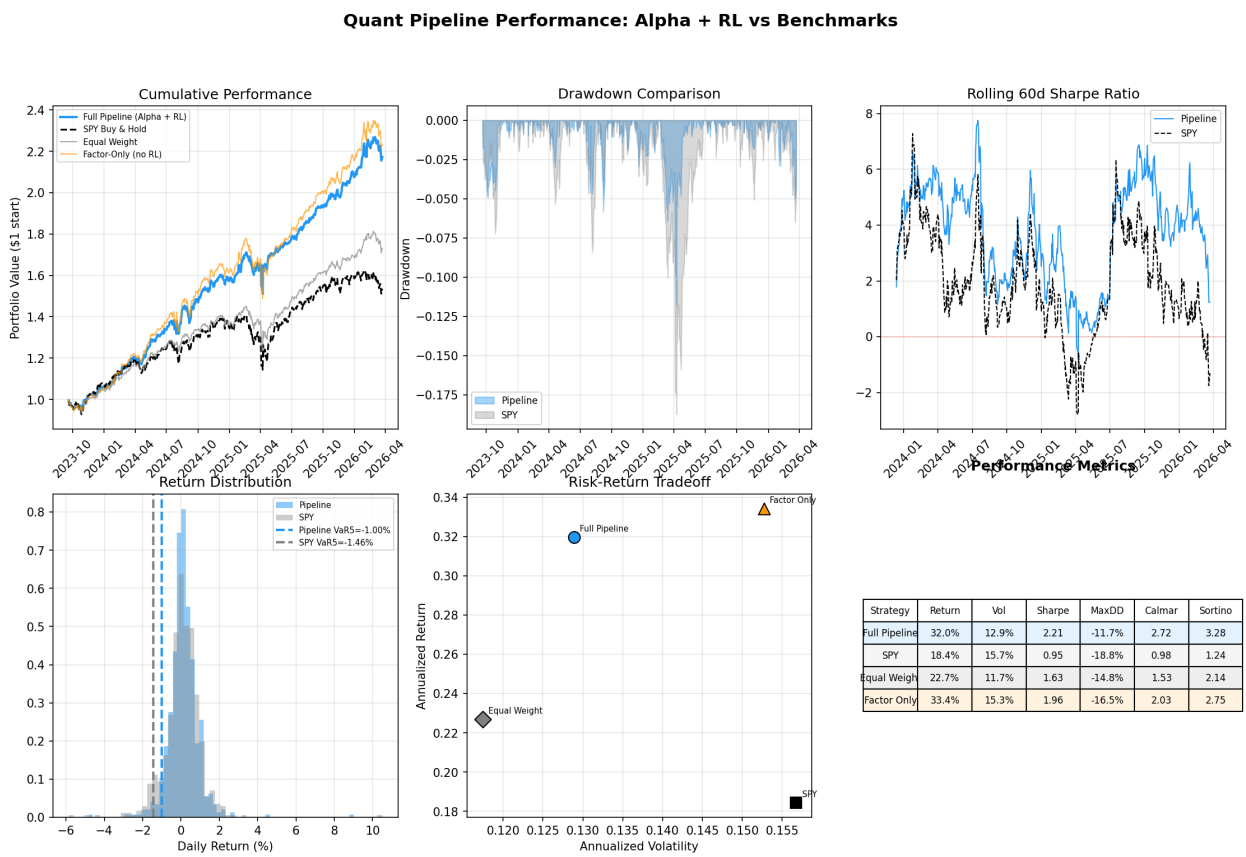
$$\text{Calmar} = \frac{\mu_{\text{ann}}}{|\text{MaxDrawdown}| + \varepsilon}.$$

8 Current Empirical Read

On the latest strong run reflected in the generated figures dated March 23, 2026, the full pipeline delivered approximately:

- annualized return: 32.0%,
- annualized volatility: 12.9%,
- Sharpe ratio: 2.21,
- maximum drawdown: -11.7%,
- Calmar ratio: 2.72.

Against the same window, the factor-only benchmark achieved slightly higher raw return but with higher volatility and a materially deeper drawdown. This is the intended shape of the system: keep the pipeline close to the factor engine in good markets while using RL to moderate exposure and protect downside.



Strategy	Return	Vol	Sharpe	MaxDD	Calmar	Sortino
Full Pipeline	32.0%	12.9%	2.21	-11.7%	2.72	3.28
SPY	18.4%	15.7%	0.95	-18.8%	0.98	1.24
Equal Weigh	22.7%	11.7%	1.63	-14.8%	1.53	2.14
Factor Only	33.4%	15.3%	1.96	-16.5%	2.03	2.75

Figure 1: Performance comparison for the full pipeline and benchmarks.

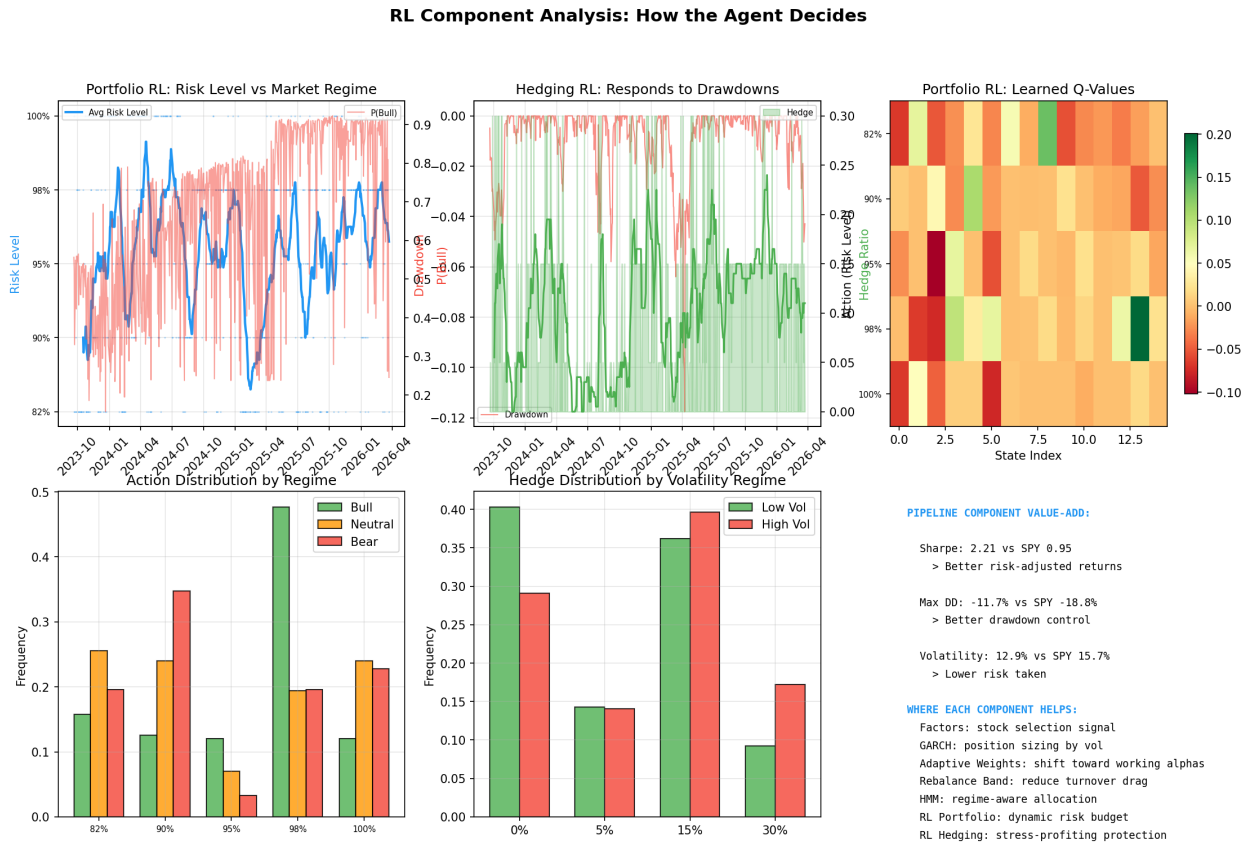


Figure 2: Behavioral diagnostics for the portfolio and hedging RL layers.

9 Limitations

The project remains research code with several important caveats:

- Macro data is not yet modeled with full vintage-awareness, so revised data can still make the backtest more optimistic than a live system.
- SEC-based quality signals are simple cross-sectional enrichments, not a full point-in-time fundamentals pipeline.
- Transaction-cost modeling is stylized and does not account for full microstructure realism.
- The hedge sleeve approximates convex protection but does not yet use live options-chain data or implied volatility surfaces.
- The test period is still only one realized market path, so robustness should be checked with subperiod analysis, parameter sweeps, and stress testing.

10 Conclusion

The main contribution of this project is not any single model family in isolation. Rather, it is the integration of several financially motivated sleeves into a coherent decision pipeline where reinforcement learning is used as a control layer instead of as a replacement for domain structure.

The modular code organization now mirrors that conceptual architecture. This makes the system easier to maintain, easier to explain, and easier to extend toward more realistic data, better point-in-time fundamentals, and richer execution and hedging models.

A Implementation Snapshot

A.1 Universe and Benchmark

Bucket	Assets
Growth	AAPL, MSFT, GOOGL, AMZN, META, NVDA
Financials / Energy	JPM, GS, XOM, CVX
Defensives	JNJ, PG, KO, PEP, WMT
Diversifiers	GLD, TLT, XLU, VNQ
Benchmark	SPY

A.2 Current RL Control Ladders

Portfolio action	Invested fraction	Active overlay fraction
State 0	82%	5%
State 1	90%	8%
State 2	95%	12%
State 3	98%	16%
State 4	100%	20%

Hedge action	Hedge ratio
State 0	0%
State 1	3%
State 2	8%
State 3	15%